

Project Synopsis: Full-Stack Attendance Portal

Prepared by: Anshuman Singh

1. Objective

The primary objective of this project was to develop a modern, secure, and efficient MERN stack application to streamline attendance, leave, and user management. The application enhances security and user experience by incorporating advanced features such as face recognition login and geofencing. It is designed to solve real-world organizational challenges and serve as a valuable asset for companies like VISON Software Solutions.

2. Modules & Key Features

The project is architected into distinct modules, each with specific functionalities:

Authentication Module

- **Secure Registration & Login:** Robust registration and login system for both Employees and Administrators, with passwords securely hashed using bcryptjs.
- **Advanced Face Recognition Login:** Allows users to register their face via webcam from their profile page and subsequently log in seamlessly without a password.
- **JWT Session Management:** Utilizes JSON Web Tokens (JWT) stored in HTTP-Only cookies for secure and persistent user sessions.

Employee Module

- **Dynamic Dashboard:** A user-friendly dashboard enabling employees to perform multiple Punch-in/out actions throughout the day.
- **Profile Management:** Users can view and update their personal details, including their name, email, and Date of Birth (DOB).
- **Leave Management:** Employees can apply for leave and track the live status (Approved/Rejected/Pending) of their applications.
- **Attendance History:** Provides a detailed, collapsible view of the user's complete attendance history, including all punch-in/out timestamps for each day.

Admin Module

- **Role-Based Access Control:** Access is restricted exclusively to users with the 'admin' role.
- **Centralized Dashboard:** A comprehensive dashboard for administrators to view all user attendance records in one place.
- **User Management:** Admins can view a complete list of users, delete users, and edit their details (name, email, DOB, role).
- **Leave Management:** Admins have the authority to approve or reject all employee leave requests.

- **Geofence Management:** Admins can create, delete, and manage multiple office locations (geofences) with specific names, coordinates, and radii, and assign them to individual users.

Security & Auditing Module

- **Geofencing:** The system validates the user's current location against their assigned (or a default) work area during punch-in/out and face login events.
- **Login History:** For every successful face login, the system logs the user's IP address, Location (Latitude/Longitude), and Device/Browser details to the database for auditing purposes.

3. Tech Stack

Category	Technology / Library
Frontend	React.js, React Router, Zustand
	face-api.js, React Bootstrap, Axios
Backend	Node.js, Express.js
Database	MongoDB, Mongoose
Authentication	JWT, bcrypt.js

4. System Architecture

The project is based on the MERN stack architecture:

1. **Frontend (React):** Manages the user interface and user interactions. It sends API requests to the backend upon user actions (e.g., button clicks).
2. **Backend API (Node.js/Express.js):** Serves as the application's "brain." It processes requests from the frontend, executes business logic (like geofence validation), and communicates with the database.
3. **Database (MongoDB):** All application data (users, attendance, leaves, etc.) is securely stored and managed using Mongoose schemas.

5. Conclusion

This project is now a fully functional application incorporating all essential features of a modern web application. The code is clean, manageable, and scalable. Future enhancements could include adding reporting/analytics features and deploying the application to cloud platforms like Vercel and Render.